From detector signals to Δ(x,Q²): the reconstruction chain of the inclusive and coherent cos 2φ measurements

2026-08-24 · companion to the cos 2φ projection report (money plots 5–7) and to plans/07 WP3 · §§1–6: the chain and the audit, quantified with `polligen.reco` (`evgen/scripts/reco_chain_figures.py`) · §7 (same day, second pass): the reconstructed-level pseudo-experiments of WP3 — money plots 5R, 7R and 6R from `polligen.recopseudo` (`money_cos2phi_reco.py` , `money_cos2phi_coherent_reco.py` ; 26 tests) · revised 2026-08-25 against the references in `refs/` : the ePIC Roman-Pot slot geometry and beam-effect resolution [17], the ZEUS bounds on the diffractive azimuthal harmonics [13,14], the exact spin-1 harmonics at finite γ [12], and the anchor's frame and convention [9]

The projection report presents the double-helicity-flip structure function Δ(x,Q²) as φ′ pseudo-data and as extracted xΔ points. This note asks the experimentalist's question behind those plots: what does the detector actually record, event by event, in the inclusive and in the coherent intact-^{6}Li measurement; how is the azimuth that carries the observable built from the reconstructed four-vectors and the known spin direction; which kinematic reconstruction assigns (x, Q²) to an event; which estimator turns spin-sorted φ′ histograms into an amplitude; and how the amplitude becomes Δ. It then audits, step by step, which of these operations the simulation performs. The audit's conclusion is that the generator-level machinery is complete and validated but that the *measurement* layer — four-vectors, frames, kinematic reconstruction, spin-state sorting, the Roman-Pot measurement — was absent; a seed of that layer now exists, and three of its first results change the analysis design.

> **Three findings that change the analysis.** (1) Three of the four inclusive sweet-spot bins of money plot 5 sit at y = 0.010–0.025, where the scattered electron carries 98–99% of the beam energy and the electron method gives δy/y = 50–120% even with a 1.2% calorimeter: x must come from the hadronic final state (Σ or Jacquet–Blondel y), which the simulation does not generate. With a 15–20% hadronic-y resolution the super-bins keep 75–83% purity and the amplitude in the reconstructed bin is within 1–4% of the true-bin value. (2) The single-fill cos 2φ′ fit — the estimator of money plots 5–7 — measures the detector's own cos 2φ′ acceptance harmonic divided by P_{zz}: a 3% harmonic aligned with the vertical spin axis (the natural symmetry axis of both the crossing angle and the Roman-Pot cutout) fakes an amplitude four times the signal. The spin-state-sorted ratio of m = ±1-rich and m = 0-rich fills cancels any acceptance bin by bin, turns relative-luminosity errors into a constant, and is 1.5× *more* precise than the single fill. (3) In the coherent channel the anchored deformation term modulates the *recoil* azimuth $\varphi_t - \varphi_S$, not the electron azimuth used in the projection; the Roman-Pot cutout fakes $\langle \cos 2\varphi_t \rangle \approx 0.5$ for a 25% aspect ratio (against a physics a2 ≈ 0.036), and the near-beam cut is angular, so the tag acceptance is 67% / 9% / 10^{-8} at 20.5 / 50 / 137.5 GeV/u for the same optics.

1 · What is measured

Both measurements share the trigger object — a scattered electron in the central detector — and the spin bookkeeping of the ion beam. The coherent measurement adds the intact ^{6}Li in the far-forward spectrometers and an exclusivity selection. The observable in both cases is not a cross section but a *Fourier coefficient of a ratio* of spin-sorted azimuthal distributions; Δ follows from that coefficient through kinematic factors and the unpolarized structure functions of the same data set.

RECORDED PER EVENT	INCLUSIVE E ^{6}LI → E′ X	COHERENT E ^{6}LI → E′ X ^{6}LI(G.S.)
Beam and spin	Bunch-crossing ID → spin state of the ion bunch (source pattern: m = ±1-rich, m = 0-rich); tensor polarization P_{zz} per fill from the polarimeter (scale ≈ 3%); alignment axis n̂ = stable spin direction (vertical at IP6) with its measured tilt; luminosity per spin state.	

Central detector	e′ candidate: track (p, θ, φ at the vertex, charge) matched to an EMCal cluster (E, position); E/p and shower shape for electron ID; the hadronic final state (all other tracks and clusters) for $\Sigma = \Sigma_h(E_h - p_{z,h})$; vertex.	The same, plus the diffractive system X (its mass M_X and the rapidity gap) — the only handle on x_P, since the recoil momentum is unresolved.
Far forward	Not required (breakup fragments of ^{6}Li are mostly beam-blind at IP6; plans/00).	Roman-Pot track at beam rigidity (|R − 1| < 0.05) with transverse displacement above the 10σ envelope → (θ_x, θ_y) at the IP via the optics transport → p_T, φ_t, t; vetoes in OMD (p, ^{3}He), ZDC (n), B0 EMCal (de-excitation γ); Z-identification if available (open question #19).
Observable	cos 2φ′ coefficient of the ratio of spin-sorted $N_f(\varphi')$ in (x, Q^2) bins, $\varphi' = \varphi_e - \varphi_S$	cos 2α, cos 2β and cos(α+β) coefficients of the same ratio in (x_P, t, Q^2) bins, $\alpha = \varphi_e - \varphi_S$, $\beta = \varphi_t - \varphi_S$

2 · Angles and frames

For an unpolarized electron on a spin-1 target whose alignment axis has polar angle θ_S and azimuth φ relative to the virtual photon and the lepton plane, the Hoodbhoy–Jaffe–Manohar cross section for the projection state m reads [1,2]

$$\frac{d\sigma^{(m)}}{dx\,dy\,d\phi} = \frac{2y\alpha^2}{Q^2}\Big[F_1 + \tfrac{2}{3}a_m b_1 + \tfrac{1-y}{xy^2}\big(F_2 + \tfrac{2}{3}a_m b_2\big) - \tfrac{1-y}{y^2}\,c_m \sin^2\theta_S\,\Delta(x,Q^2)\cos 2\phi\Big], \qquad c_m = 3m^2 - 2$$

and a fill with populations p_m along a transverse axis (θ_S = 90°) has the φ′ distribution 1 + a2 cos 2φ′ with

$$a_2 = -P_{zz}\frac{1-y}{y^2}\frac{\Delta}{D_\phi}, \qquad D_\phi = F_1 + \frac{1-y}{xy^2}F_2, \qquad \hat{A} \equiv \frac{a_2}{P_{zz}}, \qquad P_{zz} = \sum_m p_m c_m.$$

Which angle. The φ of the master formula is defined in a frame in which the target momentum P and the virtual photon q are collinear; it is the azimuth of the transverse spin about q measured from the lepton plane. The covariant definition used for every transverse-spin azimuth in SIDIS and exclusive analyses [3,4] applies verbatim to the spin four-vector S (and, with P′ in place of S, to the recoil azimuth of §4):

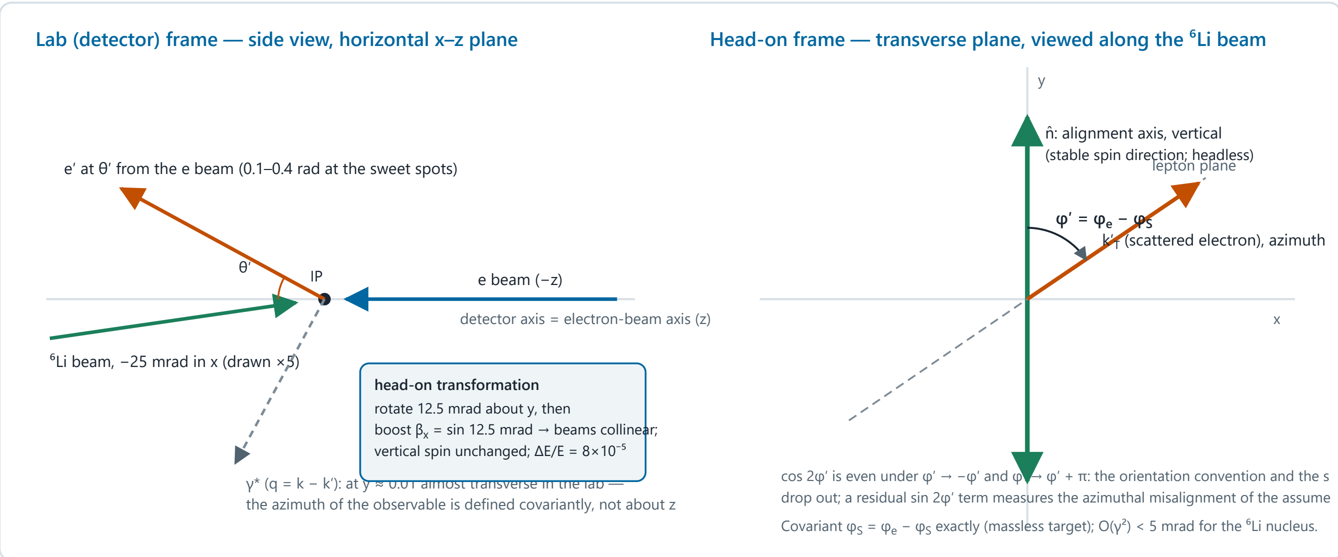
$$\cos\phi_S = -\frac{l_\perp\cdot S_\perp}{|l_\perp||S_\perp|}, \qquad \sin\phi_S = -\frac{\varepsilon_{\mu\nu\rho\sigma}l^\mu S^\nu P^\rho q^\sigma}{(P\cdot q)\sqrt{1+\gamma^2}\,|l_\perp||S_\perp|}, \qquad v_\perp^\mu = g_\perp^{\mu\nu}v_\nu, \quad \gamma^2 = \frac{M^2Q^2}{(P\cdot q)^2}$$

with $g_\perp$ the projector orthogonal to P and q [3]. Evaluated in the head-on frame — ion along +z, electron along −z, the frame of the whole fast simulation — with a spin axis transverse to the beam at lab azimuth φ_S and the scattered electron at lab azimuth φ_e, this reduces *exactly* to $\varphi' = \varphi_e - \varphi_S$ for a massless target, and to the same with corrections of order γ^2 (below 5 mrad for $\gamma^2 \le 0.02$, i.e. everywhere in the sweet spots) for the massive nucleus (`reco.azimuth_wrt_lepton_plane`, verified against an explicit boost-and-rotate construction of the collinear frame to 10^{-13}). The lab-angle shortcut is therefore legitimate for the inclusive measurement; the analysis should nevertheless evaluate the covariant formula from the reconstructed four-vectors, because it is exact, costs nothing, and is the same code that the coherent channel needs.

Which frame. The beams cross at 25 mrad in the horizontal plane at IP6. The ePIC axis coincides with the electron beam, and the ion beam enters at −25 mrad in x (the convention of the far-forward gun scan in `tools/fullsim`). The head-on frame is reached by a rotation of 12.5 mrad about y followed by a boost β_x = sin 12.5 mrad; energies change by 8×10^{-5} and the vertical spin direction is untouched. For the scattered electron the boost and the rotation cancel at first order because the detector axis is the electron axis: the residual distortion of its azimuth about the detector axis is an odd-harmonic effect of 0.7×10^{-3} (θ′ = 107 mrad) to 2.4×10^{-3} (383 mrad) in cos φ, with the cos 2φ harmonic untouched to 10^{-17} and the response of a true cos 2φ term equal to one within 10^{-5}. The transformation is applied anyway (it is the standard first step of every ePIC inclusive analysis and it is exact), but the crossing angle is not a source of a fake cos 2φ′ for the electron. It *is* one for the recoil (§4).

Orientation and the axis. The tensor alignment axis is headless — the m = +1 and m = −1 populations enter through $c_m = 3m^2 - 2$ symmetrically — and cos 2φ′ is even under φ′ → −φ′ and φ′ → φ′ + π, so neither the orientation convention of φ nor the sign of $\hat{n}$ can change the result. What matters is the axis direction modulo π: a misalignment δ of the assumed azimuth rotates the modulation, $\cos 2(\varphi' - \delta) = \cos 2\delta \cos 2\varphi' + \sin 2\delta \sin 2\varphi'$, so the fit should carry a sin 2φ′ term; since the only physics harmonic in the tensor ratio is cos 2φ′ (the m-symmetric fills set $a_1 = 0$, and the b_1-type constant is φ′-independent), the sin 2φ′ coefficient measures δ from the data themselves — a self-calibration of the spin axis at the precision of the modulation itself. A polar tilt ε of the axis out of the transverse plane dilutes by $\sin^2\theta_S = 1 - \varepsilon^2$ (0.8% at 5°).

Harmonics from the tensor structure functions at finite γ. The exact spin-1 decomposition for unpolarized electrons (Cosyn et al. [12], Eq. 10) carries, besides the leading-twist Δ term, tensor-polarized pieces with the same azimuthal signatures: the combination of b_2, b_3, b_4 multiplying $\cos\varphi_{TL}$ is of order γ (their Eq. 17d), and the combination of b_2, b_3 multiplying $\cos 2\varphi_{TT}$ is of order γ^2 (Eq. 17e), with $\gamma = 2Mx/Q$. At the sweet spots $\gamma^2 \le 0.023$, so the b-type leakage into the cos 2φ′ amplitude is of order $\gamma^2(b_1/F_1)/6 \approx 4\times10^{-4}$ for a HERMES-like $b_1/F_1 \approx 0.1$ — at most 3% of the Δ amplitude, at the high-x, low-Q^2 corner only, and computable from the same data's A_{zz}; the O(γ) cos φ′ term is orthogonal to the cos 2φ′ fit at full coverage and becomes a by-product of the ratio fit. (Their paper drops Δ altogether, its subject being b_1; for the axis along q the T_{LT} and T_{TT} parameters vanish, their Eq. 20, which is why that axis is the clean choice for A_{zz} and the transverse axis the only choice for Δ.)



Schematic 1 — frames and the azimuth of the observable. Left: the detector frame with the 25 mrad crossing angle; the scattered electron is measured at (E′, θ′, φ_e) about the electron-beam axis, and the virtual photon is far from the beam axis at low y. Right: in the head-on frame the alignment axis is vertical and the physics azimuth is the angle between the lepton plane and the axis, which the covariant formula reproduces as $\varphi_e - \varphi_S$.

3 · The inclusive chain

1. **Event selection and electron identification.** One e′ candidate with a track matched to an EMCal cluster; E/p and shower-shape cuts against π^- (photoproduction) and a charge-symmetric subtraction (wrong-sign tracks, positron running) for pair-symmetric background at high y. Efficiency $\varepsilon_{eID}(\eta)$ = 0.70–0.95 in the ATHENA/ECCE anchors used by the money_delta study; no official ePIC curve exists.
2. **Scattered-electron kinematics.** E′ from the calorimeter (2%/√E ⊕ 1%, i.e. 1.2% at E′ ≈ E_e) or the tracker (3% at η ≈ −3 in the tracking-only table), θ′ and φ_e from the track at the vertex (1–3 mrad), transformed to the head-on frame. The four-vectors k, k′, P and the spin four-vector $S = (0, \hat{n})$ enter the covariant azimuth.
3. **Kinematic reconstruction.** The electron method,

$$Q_e^2 = 4E_eE^{'}\sin^2\frac{\theta^{'}}{2}, \qquad y_e = 1 - \frac{E^{'}}{E_e}\cos^2\frac{\theta^{'}}{2}, \qquad x_e = \frac{Q_e^2}{s\,y_e}, \qquad \frac{\delta y_e}{y_e} = \frac{1-y}{y}\left[\frac{\delta E^{'}}{E^{'}} \oplus \tan\frac{\theta^{'}}{2}\,\delta\theta^{'} \oplus \frac{\delta E_e}{E_e}\right]$$

has an excellent Q^2 ($\delta Q^2/Q^2 \approx \delta E'/E' \oplus \cot(\theta'/2)\delta\theta' \approx 1$–$2\%$) but loses y as $(1-y)/y$: at y = 0.01 a 1.2% energy resolution gives δy/y = 120%, and the electron-ring energy spread alone (≈ 10^{-3}) gives 10%. Table 1 shows where money plot 5 sits. At the mid-energy configuration three of the four sweet spots are at y = 0.010–0.025, the scattered electron carries 98–99% of the beam energy at η ≈ −2.9 to −2.4, and x is not measurable from e′ alone; at the top energy all four spots sit at y ≈ 0.01. The remedy is the one HERA used at low y [5,6]: y from the hadronic final state,

$$y_{JB} = \frac{\Sigma}{2E_e}, \qquad y_\Sigma = \frac{\Sigma}{\Sigma + E^{'}(1-\cos\theta_e)}, \qquad \Sigma = \sum_{h\neq e^{'}}(E_h - p_{z,h}), \qquad x = \frac{Q_e^2}{s\,y_\Sigma}\ \text{(mixed method)}$$

with θ_e measured from the ion direction. The Σ method is insensitive to collinear initial-state radiation, and both are insensitive to particles lost along the ion beam ($E - p_z \approx 0$ for forward remnants). Their resolution is set by hadronic calorimetry and by losses in the electron direction; the Yellow-Report-era studies reach y ≈ 0.01 with δy/y of order 15–30%, which is the band assumed here. The low-energy configuration moves the same (x, Q^2) to y = 0.03–0.2 (Table 1), so the energy scan is itself part of the reconstruction strategy for the x ≈ 0.1 bins.

SWEET SPOT	X, Q^2 [GEV²]	E(5)×⁶LI(20.5): Y	E(10)×⁶LI(50): Y, E′ [GEV], Θ′ [MRAD], Η	E(18)×⁶LI(137.5): Y	ΔY/Y, E′ ALONE (1.2%)
1	0.056, 1.14	0.031	0.010, 9.93, 107, -2.93	0.011 (x = 0.014)	1.2
2	0.022, 1.14	0.078	0.025, 9.78, 108, -2.92	0.026 (x = 0.0056)	0.47
3	0.141, 3.13	0.054	0.011, 9.97, 177, -2.42	0.011 (x = 0.028)	1.1
4	0.141, 14.3	0.196	0.051, 9.85, 383, -1.64	0.010	0.23

Table 1 — the four sweet-spot super-bins of money plot 5 (selected per Q^2 band from the significance map at e(10)×⁶Li(50)) and where the same bins sit at the other two energies; the top-energy spots are chosen by the same rule and land at different x.

4. **The azimuth.** $\varphi' = \varphi_e - \varphi_S$ from the covariant formula (§2), with φ_S the alignment-axis azimuth of that bunch from the spin bookkeeping. The angular resolution of the track (1–3 mrad) dilutes cos 2φ′ by $\exp(-2\sigma_\varphi^2) \approx 1 - 2\times10^{-5}$: negligible.
5. **Binning in reconstructed (x, Q²).** Figure 1(c) quantifies the migration for the mixed method with the sampler's own events: with $\delta y_{had}/y$ = 15% (20%, 30%) the 3×3 super-bins keep a purity of 0.81–0.83 (0.74–0.77, 0.61–0.65) and the amplitude seen in the reconstructed bin is 0.98–0.99 (0.96–0.99, 0.92–0.98) of the true-bin value, because A(x) varies slowly across the bin. With the electron method alone the low-y spots collapse to purities of 0.41–0.56 with 26–53% of the events surviving in the bin. The standard treatment — a migration matrix from the same simulation and either an unfolding or model bin-centering — is the same as for any structure-function measurement; what is new is only that the correction must be built on the hadronic-method resolution.
6. **Spin-state sorting and the estimator.** The events of each (x, Q²) bin are histogrammed in φ′ separately for each spin state f of the bunch pattern, $N_f(\varphi')$. Any smooth φ′-dependent efficiency ε(φ′) multiplies every N_f identically, so the bin-wise ratio

$$R(\phi^{'}) = \frac{\sum_f w_f Y_f(\phi^{'})}{\sum_f Y_f(\phi^{'})}, \qquad Y_f = \frac{N_f}{L_f}, \qquad w_f = P_f - \bar{P}, \qquad R(\phi^{'}) \simeq \sigma_P^2\left[\kappa + \hat{A}\cos 2\phi^{'}\right], \qquad \sigma_P^2 = \frac{\sum_f L_f(P_f - \bar{P})^2}{\sum_f L_f}$$

is free of ε(φ′) exactly, and a relative-luminosity error δ_f enters it only as the φ′-independent constant $\Sigma w_f\delta_f/F$, orthogonal to the harmonic (second order: $\delta \times \hat{A} \approx 10^{-6}$). A weighted least-squares fit of R(φ′) = c + d cos 2φ′ (+ e sin 2φ′) gives $\hat{A} = d/\sigma_P^2$ after the finite-bin dilution sin w/w is divided out, and $\kappa = c/\sigma_P^2$ is the transverse tensor asymmetry of the b₁ sector, a by-product. For two fill types with equal luminosity the statistical error is

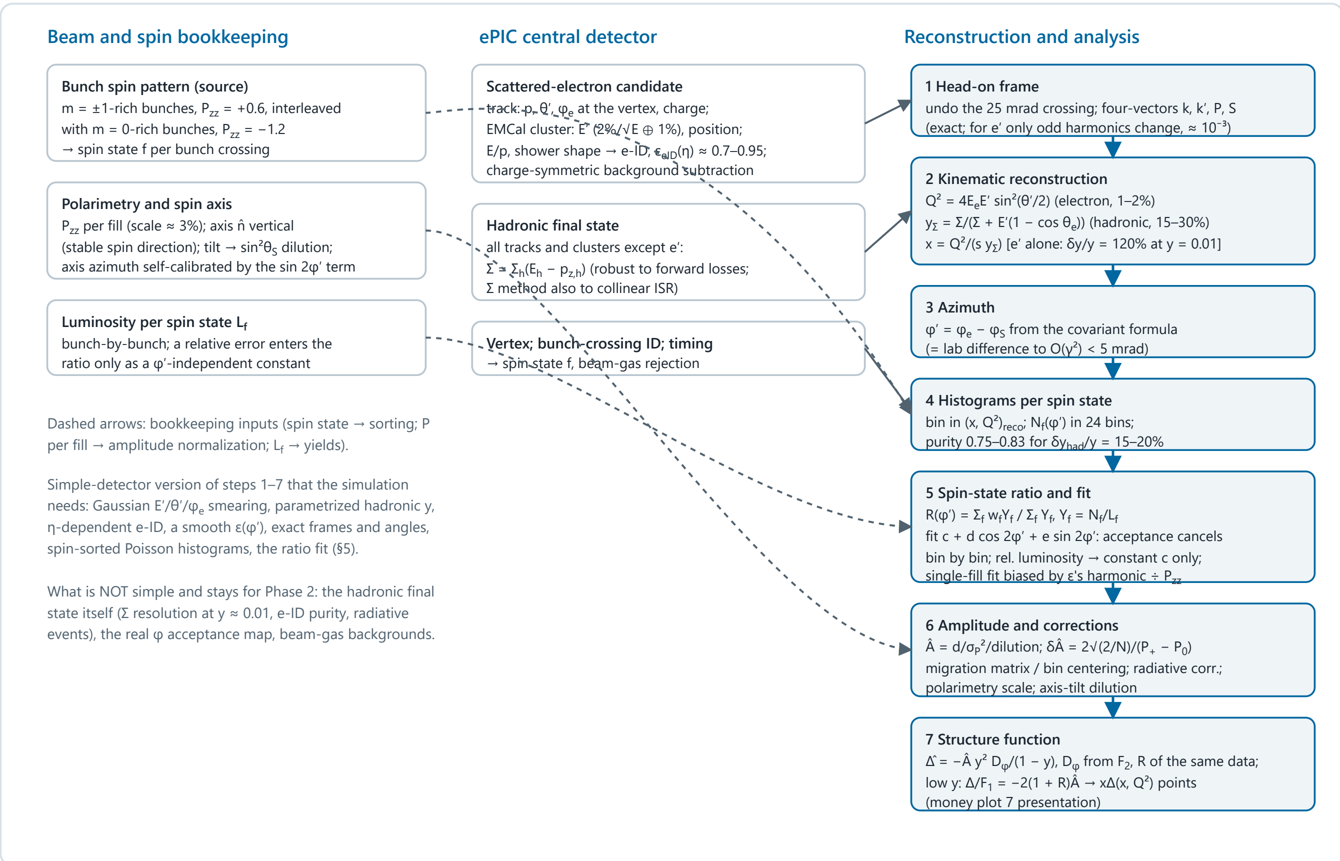
$$\delta\hat{A} = \frac{2}{P_+ - P_0}\sqrt{\frac{2}{N}}\ \text{(two fills)} \qquad \text{vs} \qquad \delta\hat{A} = \frac{1}{P_{zz}}\sqrt{\frac{2}{N}}\ \text{(single fill)}$$

so that m = ±1-rich bunches at $P_+ = +0.6$ alternated with m = 0-rich bunches at $P_0 = -1.2$ (the same source purity gives $-2P_+$, the pattern of the A_{zz} run plan) are 1.5× more precise than a single $P_{zz} = 0.6$ fill with the whole luminosity; an unpolarized reference fill ($P_0 = 0$) would instead cost a factor two. Figure 1(d) shows the pseudo-experiments: under an efficiency $\varepsilon = 1 + 0.03\cos 2\varphi' + 0.02\cos\varphi'$ the single-fill fit returns $\hat{A} = 0.062$ for a true 0.0118 — the acceptance harmonic divided by P_{zz} — while the ratio is unbiased with the analytic error. The projection report's statement that the modulation is "self-normalized within a single fill" holds for relative luminosity but not for acceptance; since a percent-level cos 2φ harmonic aligned with the vertical axis is expected (the crossing-angle plane is horizontal, module boundaries and dead areas are not azimuthally random, and the Roman-Pot cutout is a rectangle), the spin-state-sorted ratio is the estimator of the measurement.

7. **Amplitude to Δ.** With F2 and $R = \sigma_L/\sigma_T$ of ^{6}Li per nucleon measured in the same data (the unpolarized cross section of the same sample, with its own radiative corrections) or taken from nuclear-PDF fits,

$$\hat{\Delta}(x,Q^2) = -\hat{A}\frac{y^2 D_\phi}{1-y} \;\to\; -\hat{A}\frac{F_2}{x} = -2(1+R)F_1\hat{A} \quad (y \ll 1), \qquad \text{i.e.}\;\; \frac{\Delta}{F_1} = -2(1+R)\hat{A}$$

At the sweet spots the conversion is therefore insensitive to y — the $(1-y)/y^2$ factor that makes low y attractive for the amplitude cancels against D_φ — and Δ/F1 is essentially a direct ratio measurement; the y mis-measurement of the electron method affects the x label of the point, not the amplitude. Remaining corrections: the tensor rate shift $(2/3)a_m(b1 + \ldots)$ in D_φ, at the 10^{-3} level and measured by κ; target-mass terms of order γ^2; radiative corrections, uncalculated for tensor observables (plans/07 WP4); the per-nucleon and dilution conventions, which are interpretation rather than reconstruction; and the polarimetry scale $\delta P_{zz}/P_{zz} \approx 3\%$, common with A_{zz}.



Schematic 2 — the inclusive reconstruction chain. Left and middle: what is recorded; right: the seven reconstruction and analysis steps of §3 with the quantity each produces and its dominant limitation. The measurement's observable is the cos 2φ′ coefficient of the spin-state ratio (step 5), not the modulation of a single histogram.

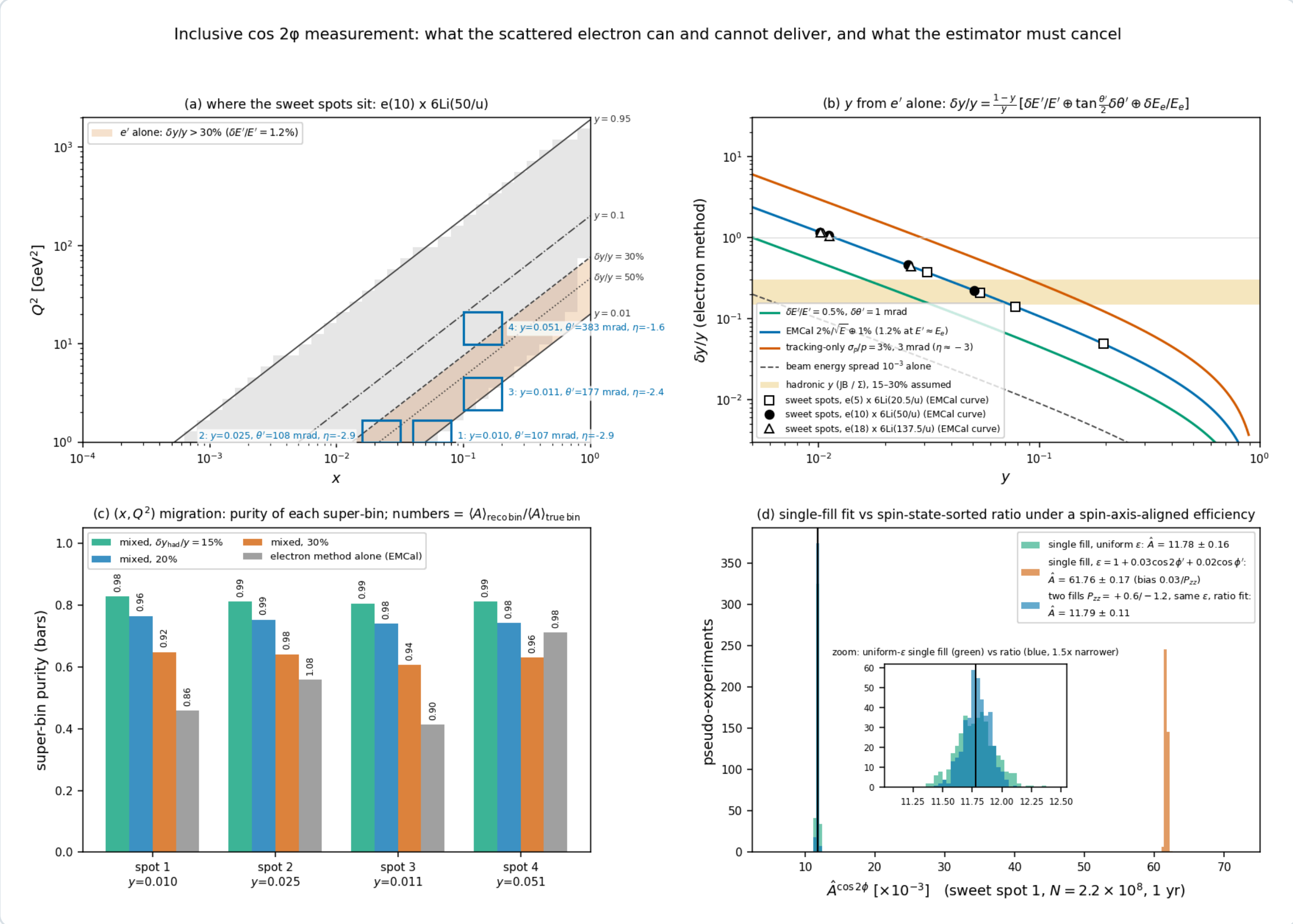


Figure 1 — the inclusive measurement quantified (e10 × ^{6}Li50, `reco_chain_figures.py`). **(a)** The analysis acceptance with iso-y lines and the four sweet-spot super-bins of money plot 5; the shaded band is where the electron alone gives δy/y > 30% for a 1.2% energy resolution: spots 1–3 lie inside it, spot 4 just above. **(b)** The electron-method δy/y versus y for calorimeter, tracking, and beam-energy-spread inputs, with the sweet spots of the three energies placed on the calorimeter curve and the assumed 15–30% hadronic-method band. **(c)** Purity of each super-bin after smearing the sampler's events with the mixed method for three hadronic-y resolutions and with the electron method alone; the numbers give the ratio of the amplitude averaged over the events landing in the reconstructed bin to that of the true bin. **(d)** Four hundred pseudo-experiments of the spot-1 super-bin at the 1-year statistics: the single-fill fit with a uniform efficiency (green) recovers the truth, the same fit under ε = 1 + 0.03 cos 2φ′ + 0.02 cos φ′ (vermillion) is biased by $0.03/P_{zz}$, and the two-fill ratio fit under the same efficiency (blue) is unbiased and 1.5× more precise (inset).

4 · The coherent chain

The coherent measurement keeps steps 1–7 for the electron and adds the recoil. Its kinematics is fixed by two facts of the projection report: the recoil keeps the beam rigidity (R = 1.000 for A/Z = 2, and the momentum loss $x_P < 0.01$ is below the Roman-Pot momentum resolution), so the only measured recoil quantity is its transverse displacement from the beam orbit; and coherence confines the sample to $x \lesssim 0.01$, i.e. to $y \gtrsim 0.05$ at the mid energy (median $x = 3.5\times10^{-3}$, $y \approx 0.2$), where the electron method is adequate and the low-y problem of §3 does not arise.

1. **Roman-Pot track.** Hits in the two stations (AC-LGAD, ≈ 30 ps timing for bunch association) form a local track; the optics transfer matrices referenced to the beam orbit convert it to the angle pair (θ_x, θ_y) at the IP relative to the ion axis, so the crossing angle and the reference orbit are removed by construction, and the vertex from the central detector supplies the position term. The momentum fraction x_L is set to 1 (unresolved), hence $p_T = A\, p_u \sqrt{(\theta_x^2 + \theta_y^2)}$ and $\varphi_t = \mathrm{atan2}(\theta_y, \theta_x)$. The resolution is dominated by the angular divergence of the bunch itself, $\delta p_T = A\, p_u \sigma_\theta = 22$ MeV at 50 GeV/u for $\sigma_\theta = 73$ μrad, giving $\sigma(\varphi_t) = 0.09$ rad at $p_T = 0.25$ GeV and a cos $2\varphi_t$ dilution of 1.5% (Figure 2d). The ePIC far-forward simulation with all beam effects (angular divergence, crossing angle, crab rotation and vertex smearing) gives $\Delta p_T/p_T \approx 0.09$ at 0.45 GeV and 0.04 at 1 GeV for 275 GeV protons, i.e. $\Delta p_T \approx 40$ MeV, with the detector alone contributing

≤ 1.5% — "beam effects the dominant source of momentum smearing" [17]; ZEUS's leading-proton spectrometer was limited the same way ($\sigma(t)/t = 0.14$ GeV/$\sqrt{|t|}$, Φ resolution 0.2 rad) [13].

2. **Momentum transfer.** Exactly, with $x_L = 1 - x_P$ the plus-momentum fraction,

$$t = -\frac{p_T^2 + M^2 x_P^2}{1-x_P} \simeq -p_T^2, \qquad t_{\min} = \frac{M^2 x_P^2}{1-x_P} = 3.2\times10^{-3}\ \mathrm{GeV}^2\ (x_P = 0.01), \qquad x_P = \frac{M_X^2 + Q^2 - t}{W^2 + Q^2 - M^2}$$

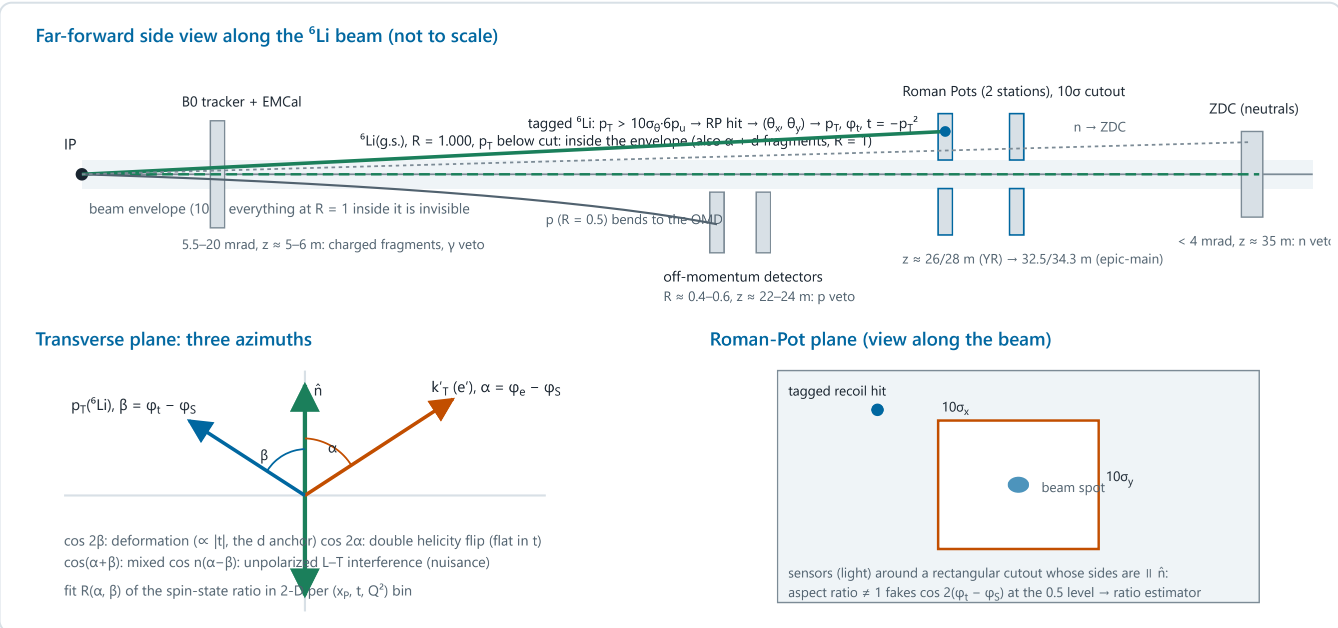
so the Roman Pots give t, and x_P (or $\beta = x_A/x_P$) must come from the diffractive mass M_X of the central detector — a coarse variable (a few x_P bins), sufficient for the x_P-dependence test of the two-component fit. The covariant recoil azimuth differs from the lab difference $\varphi_e - \varphi_t$ by less than 0.2 mrad at these kinematics, so here too the lab shortcut is safe.

3. **Exclusivity selection.** A beam-rigidity Roman-Pot track above the envelope; no OMD track (p at R = 0.5, ^{3}He at 0.75 in the main window), no ZDC neutron, no B0 EMCal photon (the 3.56 MeV state's de-excitation γ boosted to 0.1–0.4 GeV); a rapidity gap between X and the beam; Z-identification of the track if the pots resolve Z = 3 from 2 and 1 (open question #19), otherwise a two-component fit of the |t| spectrum against the ten-times-flatter α + d breakup slope; timing and vertex association against beam-gas coincidences. Because the incoherent cross section is m-state-blind, residual breakup dilutes the tensor ratio without faking it; its fraction is measured in the same data from the |t| fit.
4. **Three azimuths, two harmonics.** With the recoil measured there are two physical azimuths besides the spin: $\alpha = \varphi_e - \varphi_S$ (lepton plane to axis) and $\beta = \varphi_t - \varphi_S$ (recoil to axis), and $\alpha - \beta = \varphi_e - \varphi_t$ is the lepton-plane–recoil angle of unpolarized diffractive DIS. The rank-2 tensor polarization enters only through $e^{\pm 2i\varphi_S}$, so at leading power in the momentum transfer the spin-sorted distribution has the structure

$$N_f(\alpha,\beta) \propto \varepsilon(\alpha,\beta)[1 + u_1\cos(\alpha-\beta) + u_2\cos 2(\alpha-\beta) + P_f(\kappa + a_e\cos 2\alpha + a_t\cos 2\beta + a_m\cos(\alpha+\beta))]$$

The deformation term of the polarized-deuteron anchor — the modulation of the diffractive slope by the aligned gluon density, $a_2(t) \propto |t|$, digitized in `coherent.MANTYSAARI_A2_DEUTERON` — is a_t: it lives in the *recoil* azimuth β, requires no photon helicity flip, and vanishes as t → 0 (the anchor defines Φ as the angle between the vector-meson momentum and the polarization axis in the γ*d c.m. frame and notes that at $Q^2 > 0$ the transformation from the lab "would mix the polarization states" [9] — the O(γ) mixing that the covariant definitions of §2 absorb). The double-helicity-flip term — the coherent counterpart of Δ, an interference of the two transverse photon helicities that needs the linear polarization of the photon and hence the lepton plane — is a_e: it lives in α and is flat in t at leading power. The two mechanisms of the projection report are thus separated not only by their t-dependence but by the azimuth they modulate, which is a stronger discriminant; the mixed term a_m and the unpolarized $u_{1,2}$ are nuisance harmonics. The latter are the L–T and T–T′ interference terms of diffractive DIS with a detected recoil, whose y-dependence is $u_1 = (2-y)\sqrt{(1-y)}\, A_{LT}/[2(1-y)+y^2]$ and $u_2 = 2(1-y)\, A_{TT}/[2(1-y)+y^2]$ [14]; ZEUS measured them with the leading-proton spectrometer as $A_{LT} = -0.036 \pm 0.036$, $A_{TT} = -0.030 \pm 0.037$ for $x_P < 0.01$ and $+0.051 \pm 0.024$, -0.010 ± 0.024 for $0.01 < x_P < 0.1$ — consistent with zero [13] — and the pQCD LT-interference model gives $A_{LT} \approx 0.03$ at our $\beta \approx 0.5$, $p_T/Q \approx 0.2$ [15]; the pseudo-experiments of §7 use $u_1 = 0.05$, $u_2 = 0.02$, at the edge of the ZEUS bounds. The money-plot-6 pseudo-data place the anchored $\langle a_2 \rangle$ on an axis labelled $\varphi' = \varphi - \varphi_S$ of the electron, which is the α axis: numerically harmless for a statistical projection, but the measurement must histogram (α, β) in two dimensions per (x_P, t, Q^2) bin and extract a_t and a_e from the spin-state ratio R(α, β); in that ratio u(α − β) survives in the denominator and feeds the pure harmonics only at second order ($u_1^2 \approx 1\%$) but the mixed term at first order ($a_t u_1/2$), which fixes the fit model. (One normalization caveat to settle at the source: whether the anchor's a_n are defined with $1 + 2\Sigma a_n \cos n\Phi$ or with $1 + \Sigma a_n \cos n\Phi$ — a factor two on the digitized amplitudes that the projection code treats as the latter.)

5. **Acceptance geometry and the estimator.** The Roman-Pot acceptance is not a circular p_T cut. The ePIC pots are 25.6 × 12.8 cm AC-LGAD planes surrounding a *horizontal slot* through which the beam, its momentum spread and its dispersion pass [17]: the cutout is wide in x and tight in y (the 10σ vertical beam size), a rectangle with $c_x \gg c_y$ whose sides are again horizontal and vertical — parallel and perpendicular to the spin axis. Anisotropy is the norm, not the exception: HERA's proton beam had a transverse-momentum spread at the interaction point of 45 MeV horizontally and 100 MeV vertically [13]. For the exponential recoil spectrum the tagged sample's azimuthal distribution is $\varepsilon(\varphi_t) = \exp[-B\,\rho(\varphi_t)^2]$ with ρ the cutout boundary, and its $\langle\cos 2\varphi_t\rangle$ (about the horizontal) is 0 for a square, +0.49 for an aspect ratio $c_y/c_x = 1.25$, +0.71 for 1.5, and −0.5 to −0.8 for the slot-like ratios 0.3–0.6 (Figure 2b) — ten to twenty times the anchored $a_t \approx 0.036$, with the same symmetry axis (the sign flips when the axis is referred to the vertical spin direction). A square still carries $\langle\cos 4\varphi_t\rangle = 0.34$, which mixes into cos 2β at second order through the physics term. No single-fill fit can survive this; the spin-state ratio of §3 cancels it exactly, bin by bin in (α, β), and the same m = ±1-rich / m = 0-rich pattern serves both measurements.
6. **The near-beam cut is angular.** The documented 0.20 GeV (high-acceptance) and ≈ 0.41–0.45 GeV (high-divergence) cuts are ten beam divergences times the 275 GeV proton momentum, $\sigma_\theta \approx 73$ and 149 μrad. The same envelope applied to a ^{6}Li nucleus of momentum $6p_u$ is $p_T^{cut} = 10\sigma_\theta \cdot 6p_u$: 0.09, 0.22 and 0.60 GeV at 20.5, 50 and 137.5 GeV/u, giving tag acceptances of 67%, 9% and 1.5×10^{-8} for $B = 50\ \mathrm{GeV}^{-2}$ (Figure 2c). The projection's constant 0.20 GeV cut (13.5%) is close to the mid-energy value and far from the others; whether the lithium divergence at a given energy equals the proton's is precisely the undocumented optics question of plans/04 #11, but the scaling with $A\,p_u$ is kinematics, not optics, and it makes the low-energy configuration the natural home of the coherent program.
7. **Resolution and unfolding.** The divergence smearing moves the effective slope from B to $1/(1/B + 2\delta p_T^2)$ (47.7 for the numbers above) and smears |t| by $2p_T\delta p_T \approx 0.01\ \mathrm{GeV}^2$ — small against the 0.04–0.2 GeV^2 window (Figure 2d); the t-binning migration and the acceptance edge are unfolded with the same emulation, and t_{min} is a known bias of $3\times10^{-3}\ \mathrm{GeV}^2$ at $x_P = 0.01$ once x_P is known from M_X.



Schematic 3 — the coherent channel. Top: the far-forward measurement of the intact recoil, visible only when its transverse momentum carries it outside the beam envelope at the Roman Pots, and the vetoes that define exclusivity. Bottom left: the two physical azimuths relative to the alignment axis and the harmonic each mechanism modulates. Bottom right: the rectangular cutout of the pots, whose symmetry axes coincide with the spin axis.

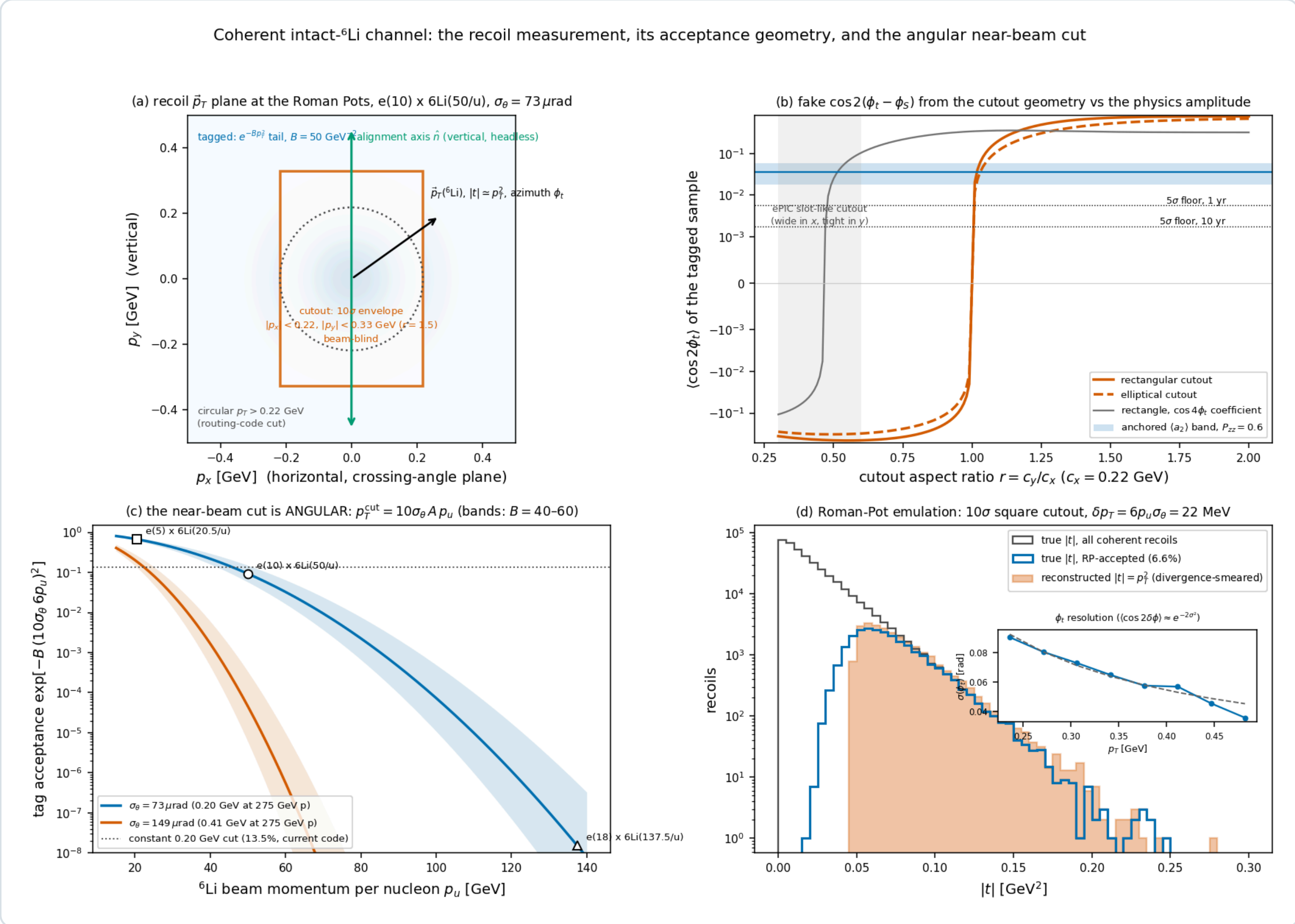


Figure 2 — the coherent measurement quantified. (a) The recoil transverse-momentum plane: the exponential coherent spectrum, the rectangular 10σ cutout (aspect ratio 1.5 drawn) against the circular cut of the routing code, and the vertical alignment axis. **(b)** ⟨cos 2φ_t⟩ of the tagged sample versus the cutout aspect ratio for rectangular and elliptical cutouts, with the cos 4φ_t coefficient of the square, the anchored deformation band and the 5σ statistical floors of the 1- and 10-year programs. **(c)** Tag acceptance versus beam momentum per nucleon for the two proton-derived divergences, with the B = 40–60 GeV^{-2} bands, the three beam configurations, and the constant-cut value of the projection. **(d)** The Roman-Pot emulation (`reco.rp_measure`): true and accepted |t| spectra, the reconstructed $|t| = p_T^2$ after divergence smearing, and (inset) the φ_t resolution against $\delta p_T/p_T$.

5 · Does the simulation do this? An audit

The table walks the chain of §§3–4 and states, for each step, what an experiment does, what the repository does, and where. "Seeded" marks the operations added in this run in `evgen/polligen/reco.py` (15 tests in `tests/test_reco.py`); they are building blocks and quantified checks. The second pass of the same day composed them into the reconstructed-level pseudo-experiments of §7 (`polligen/recopseudo.py`), which closes the "missing → seeded" rows below for the inclusive analysis and for the coherent two-azimuth fit; the table is kept as the record of the state before that pass.

STEP	EXPERIMENT	REPOSITORY	STATUS
Spin bookkeeping, luminosity per state, polarimetry	Bunch pattern, L_f, P_{zz} per fill with a 3% scale	`bookkeeping.SpinCategory / RunPlan` : categories with lumi shares, relative-luminosity offsets, polarimetry smearing (not exercised in the money plots)	exists
e′ four-vector, lab frame, crossing angle, head-on frame	Track + cluster → k′ in the lab; head-on transformation	Everything is computed in the head-on frame; `kinematics.scattered_electron` gives (E′, θ, η) for acceptance only; φ_e was never constructed. Seeded: `reco.electron_fourvector` , `head_on_to_lab` , `lab_to_head_on`	partial → seeded
Electron ID and efficiency	E/p, shower shape, charge-symmetric subtraction	`money_delta_20260729.eps_eid(η)` , ATHENA/ECCE anchors (fastsim script-local); no purity model	partial

e′ resolution	Calorimeter energy, track angles	`money_delta_20260729.smear_config` : Gaussian tracking-only σ_p/p, $\sigma_\theta = \sigma_\varphi$; φ smeared but unused. Seeded: `reco.smear_electron` , `emcal_resolution` , `tracking_resolution`	partial → seeded
Kinematic reconstruction (electron / JB / Σ / mixed)	Mixed method at low y from the hadronic final state	Electron method only (20260729); no hadronic final state anywhere in polligen. Seeded: `reco.electron_method` , `electron_method_resolution` , parametrized `hadronic_y` , `mixed_method`	missing → stand-in
The physics azimuth	Covariant φ_S from k, k′, P, S	`sample._sample_state` draws φ′ directly from the modulation and adds a known φ_S; no reconstruction. Seeded: `reco.azimuth_wrt_lepton_plane` (verified against an explicit collinear-frame construction)	missing → seeded
Reconstructed-level (x, Q^2) binning and migration	Reco bins, migration matrix, unfolding	(x, Q^2) migration in 20260729 with the electron method, amplitude carried as a per-bin truth label; money plots 5–7 bin in true (x, Q^2). Seeded: the migration study of Figure 1(c)	partial
Spin-state-sorted ratio estimator for cos 2φ′	R(φ′) of m = ±1-rich vs m = 0-rich fills; fit with sin 2φ′	`transverse_tensor_plan` is a single category; `estimators.cos2phi_fit_binned` fits one histogram with boolean holes only; the thirds ratio exists for A_{zz} alone. Seeded: `reco.harmonic_ratio_fit` , `spin_state_ratio` , `expected_counts_by_fill`	missing → seeded
Full-luminosity pseudo-data	—	`sample.phi_histogram_pseudo` : exact expected counts per φ′ bin, Poisson-fluctuated; statistically exact for a binned estimator, blind to reconstruction by construction	exists (statistics only)
Amplitude → Δ	Kinematic conversion with measured F_2, R; unfolding	`money_delta_extraction.py` : model bin-centering K-factor with true kinematics; no migration matrix, model F_2	partial
Radiative corrections	Unpolarized RC on the yields; tensor RC	None (plans/07 WP4 bound planned)	missing
Coherent: recoil four-vector and t	t from the recoil, x_P from M_X	`coherent.recoil_lab` ($p_T = \sqrt{\lvert t\rvert}$, R, x_L), `sample_t` ; t never reconstructed; x_P and M_X absent (f_{coh} written against Bjorken x). Seeded: `reco.recoil_fourvector` (exact light-cone), `t_from_fourvectors`	partial → seeded
Coherent: Roman-Pot measurement	Transport, divergence smearing, rectangular cutout, angular cut	`farforward.route_charged` : scalar, φ-symmetric p_T cut, no resolution, constant 0.20 GeV at all energies. Seeded: `reco.rp_measure` , `rp_hole_acceptance` , `tag_pt_cut`	partial → seeded
Coherent: exclusivity and vetoes	OMD/ZDC/B0 vetoes, rapidity gap, Z-ID, |t| fit	`coherent.veto_table` bookkeeping only; no event-level breakup model (BeAGLE blocked on FLUKA)	missing
Coherent: two-azimuth harmonic analysis	R(α, β) per (x_P, t, Q^2) bin	Single φ′ (electron) with the anchored a_2 of the recoil azimuth; no 2-D structure	missing
Full-simulation far-forward	Geant4 acceptance, EICrecon transport	`tools/fullsim` gun scans (hit level, RP at 32.5/34.3 m); no ^{6}Li-nucleus gun, no reconstruction	partial

> **Verdict.** The simulation is a validated *truth-level* modulation machine with exact statistics: the physics kernel, the spin-density bookkeeping and the binned estimator are correct and tested, and the money plots are statistically faithful projections. What it did not contain is the measurement layer: no four-vectors, no frames, no kinematic-reconstruction method, no hadronic final state, no reconstructed azimuth, no spin-state-sorted ratio, no Roman-Pot measurement model. Two of the missing pieces matter for the headline numbers rather than for their decoration — the hadronic-method requirement at the y ≈ 0.01 sweet spots (which changes the binning and the energy strategy, not the reach) and the estimator (which changes the error by 0.67× and removes an uncontrolled systematic) — and one changes the coherent case (the angular cut and the cutout geometry). None of them touches the generator kernel; all of them belong to plans/07 WP3.

5.1 · What a minimal realistic chain looks like in polligen

With the seeded module the reconstructed-level pseudo-experiment of WP3 is a composition, not a new design. For the inclusive channel: (i) sample spin-labelled events from `InclusiveSampler` for the m = ±1-rich and m = 0-rich categories of a two-fill run plan (`bookkeeping` already supports it); (ii) build k′ with `reco.electron_fourvector` and a uniform φ_e, transform to the lab and back, smear with `smear_electron` (calorimeter energy, track angles) and weight with $\varepsilon_{eID}(\eta)$ and a smooth $\varepsilon(\varphi')$; (iii) reconstruct Q^2 with `electron_method` and x with `mixed_method` from a parametrized `hadronic_y`; (iv) compute φ' with `azimuth_wrt_lepton_plane`; (v) histogram per spin state in reconstructed bins and fit with `harmonic_ratio_fit`; (vi) correct with the migration matrix of the same sample and convert with the model F_2; (vii) close against the truth amplitudes of `effective_modulation` — the reconstructed-level analogue of the existing generator-level gate. For the coherent channel: sample t, φ_t, x_P; build P′ with `recoil_fourvector`; emulate the pots with `rp_measure` (rectangular cutout, angular cut at the beam's p_u); compute α and β covariantly; inject a_t, a_e and the unpolarized $u_{1,2}$; fit the 2-D ratio. The hadronic final state itself, the breakup backgrounds and the real acceptance maps remain Phase-2 items and are the honest limit of "simple detector effects".

6 · Recommendations

1. Re-derive money plots 5 and 7 in reconstructed (x, Q^2) bins with the mixed method and a 15–30% hadronic-y band; quote the electron-method-only variant at $y \geq 0.05$ as the fallback, and use the low-energy configuration for the $x \approx 0.1$ bins.
2. Replace the single-fill fit by the spin-state-sorted ratio with the m = ±1-rich / m = 0-rich pattern in all projections; update δA to $2\sqrt{(2/N)}/(P_+ - P_0)$ (0.67× the current values at the same source purity) and drop the "self-normalized within a single fill" wording in favour of "acceptance and relative luminosity cancel in the spin-state ratio".
3. Fit cos 2φ′ and sin 2φ′ together; report the sin 2φ′ coefficient as the spin-axis calibration.
4. In the coherent channel, present the anchored deformation term as a modulation of the recoil azimuth, add the two-azimuth harmonic fit to the analysis description, replace the constant 0.20 GeV cut by the angular cut $10\sigma_\theta \cdot 6p_u$ in `coherent.py` (curves versus σ_θ and p_u, plans/07 WP5), and state the cutout geometry as an explicit assumption to be replaced by the ePIC design.
5. Resolve the a_n normalization convention of the deuteron anchor at the source before the amplitudes are quoted again.
6. Ask the ePIC far-forward and inclusive working groups for two numbers the simulation cannot produce: the Σ-method δy/y at y = 0.01–0.05 for e + light ions, and the Roman-Pot cutout dimensions (β*, emittance, pot geometry) for the light-ion optics.

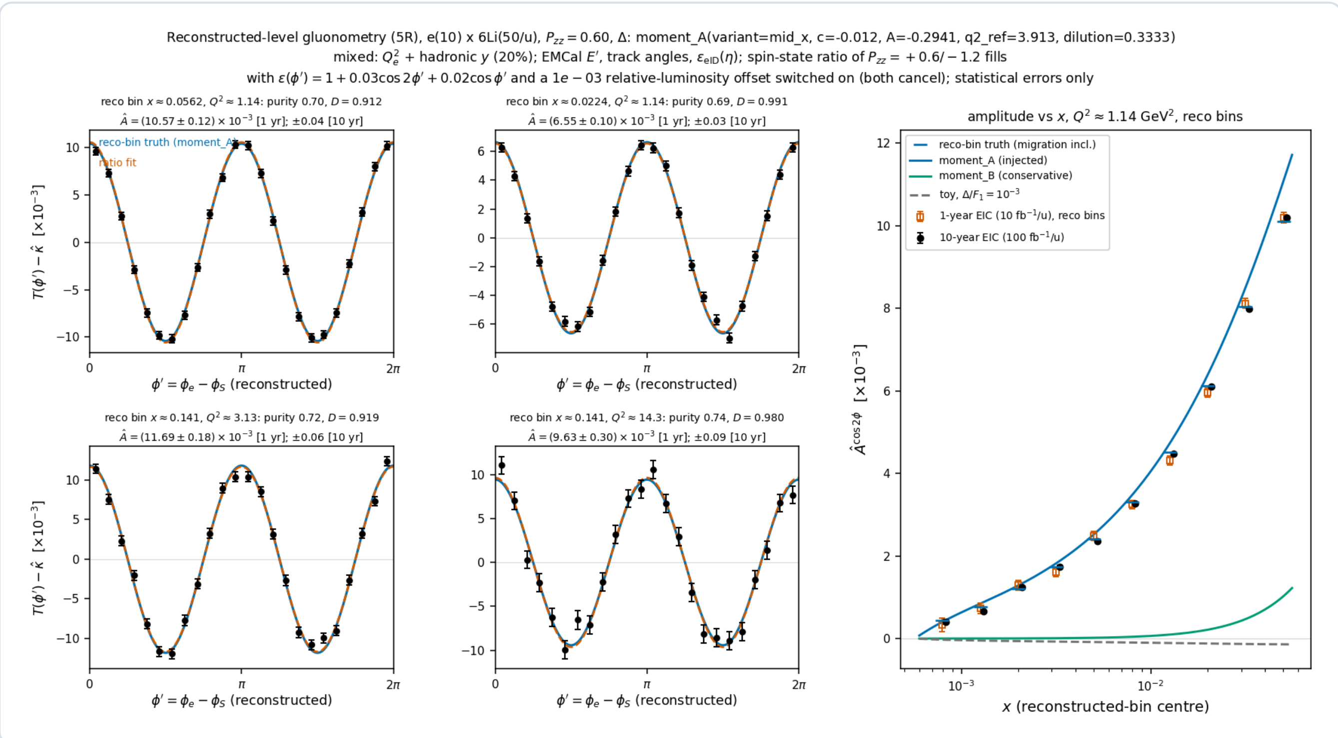
7 · Reconstructed-level closure: money plots 5R, 7R and 6R

The pieces of §5.1 are composed in `polligen/recopseudo.py`. The inclusive response is importance-sampled: every accepted sampler cell (60×45 grid, generator cuts loosened to $y \geq 0.004$, $Q^2 \geq 0.7$ GeV², $|\eta| \leq 3.8$, $E' \geq 0.3$ GeV so that events can migrate *into* the analysis window) receives 400 pseudo-events with weight $\sigma_c/400$, each carried through the lab frame (25 mrad crossing), Gaussian smearing of E′ (EMCal $2\%/\sqrt{E} \oplus 1\%$) and of the track direction (1–5 mrad by η), the head-on transformation, the electron-method Q^2, a hadronic y with 20% resolution, the mixed-method x, the analysis cuts on reconstructed quantities, $\varepsilon_{eID}(\eta)$, and the covariant φ′ on true and smeared four-vectors. The expected φ′ counts of any reconstructed bin are then exact bin integrals of $\mathrm{den}_f + P_f A \cos 2\varphi'$ summed over the events that land in it (the reconstructed-level analogue of `phi_histogram_pseudo`), Poisson- fluctuated at the 1-year and 10-year luminosities for the two fills of the m = ±1-rich / m = 0-rich pattern (`bookkeeping.tensor_flip_plan`), with the spin-axis-aligned efficiency $\varepsilon(\varphi') = 1 + 0.03 \cos 2\varphi' + 0.02 \cos \varphi'$ and a 10^{-3} relative-luminosity offset switched on. The estimator is the spin-state ratio fit of §3 with the sin 2φ′ term; Δ follows from the amplitude with the MC bin-centering factor $K = \Delta_{model}(x_c, Q^2{}_c)/A_{model,\ reco\ bin}$, which includes the migration.

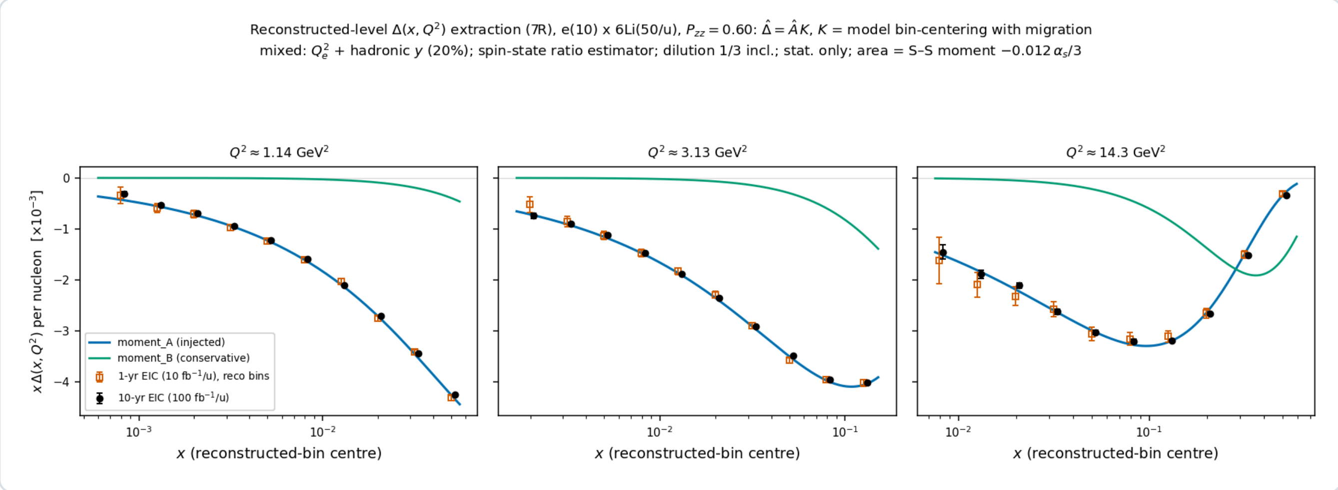
SWEET SPOT (RECO BIN)	N, 1 YR (BOTH FILLS)	PURITY	EFFICIENCY	$D = A_{\text{RECO BIN}}/A_{\text{TRUE BIN}}$	$\hat{A} \pm \Delta\hat{A}$ (1 YR) $[10^{-3}]$	RECO-BIN TRUTH	$\Delta\hat{A}$ (10 YR)	ΔA OF MONEY PLOT 5
1: x = 0.056, Q² = 1.14	1.7×10^8	0.70	0.45	0.912	10.57 ± 0.12	10.41	0.038	0.18
2: x = 0.022, Q² = 1.14	2.8×10^8	0.69	0.64	0.991	6.55 ± 0.10	6.64	0.030	0.15
3: x = 0.141, Q² = 3.13	7.6×10^7	0.72	0.40	0.919	11.69 ± 0.18	11.82	0.058	0.26
4: x = 0.141, Q² = 14.3	2.8×10^7	0.74	0.70	0.980	9.63 ± 0.30	9.43	0.095	0.45

Table 2 — money plot 5R: the four sweet-spot super-bins in reconstructed (x, Q^2) with the mixed method (20% hadronic y). Efficiency = fraction of the true-bin rate reconstructed in the bin after ε_{eID}; D = ratio of the rate-weighted model amplitude of the events in the reconstructed bin to that of the true bin. The last column is the single-fill, truth-binned error of the projection report.

Three observations. (i) The fit is unbiased at the reconstructed level: every $\hat{A}$ agrees with its reco-bin truth within the statistical error while the acceptance harmonic and the luminosity offset are on, and the sin 2φ′ coefficients are consistent with zero. (ii) The two edge bins (spots 1 and 3 at y ≈ 0.01) lose 55–60% of their events to the reconstructed y ≥ 0.01 cut and their amplitude reference shifts by 8–9% (the accepted half of the bin sits at lower true x); the analysis knows both through K, and the moment-constrained model's x-dependence is still resolved bin by bin (money plot 7R, best bins $\delta\Delta = 1.0\times10^{-3}$ at $Q^2 = 1.14$ and 0.5×10^{-3} at 3.13 GeV² in year 1, purities 0.6). (iii) The errors are *smaller* than the truth-binned single-fill projection — 0.65–0.70 of the money-plot-5 values — because the 1.5× gain of the m = 0-rich fill outweighs the reconstruction efficiency of 0.40–0.70; this is the number to carry into plans/07 §7.1, with the caveat that it presumes the source delivers m = 0-rich bunches at $|P_{zz}| = 1.2$ (the same purity as the ±1 fill at 0.6).



Money plot 5R — inclusive gluonometry at the reconstructed level (e10 × ^{6}Li50, P_{zz} = 0.6, moment_A, mixed method with 20% hadronic y, EMCal energy, track angles, $\varepsilon_{eID}(\eta)$, spin-state ratio of P_{zz} = +0.6/−1.2 fills with the acceptance harmonic and luminosity offset on). Left: the acceptance-free modulation T(φ′) − $\hat{\kappa}$ (per unit P_{zz}) from the ratio, with the reco-bin truth (blue) and the fit (vermillion); annotations give purity, D, and the 1-/10-year errors. Right: the amplitude in reconstructed x bins along the Q^2 = 1.14 GeV² slice with the reco-bin truth (blue ticks) and the model curves.

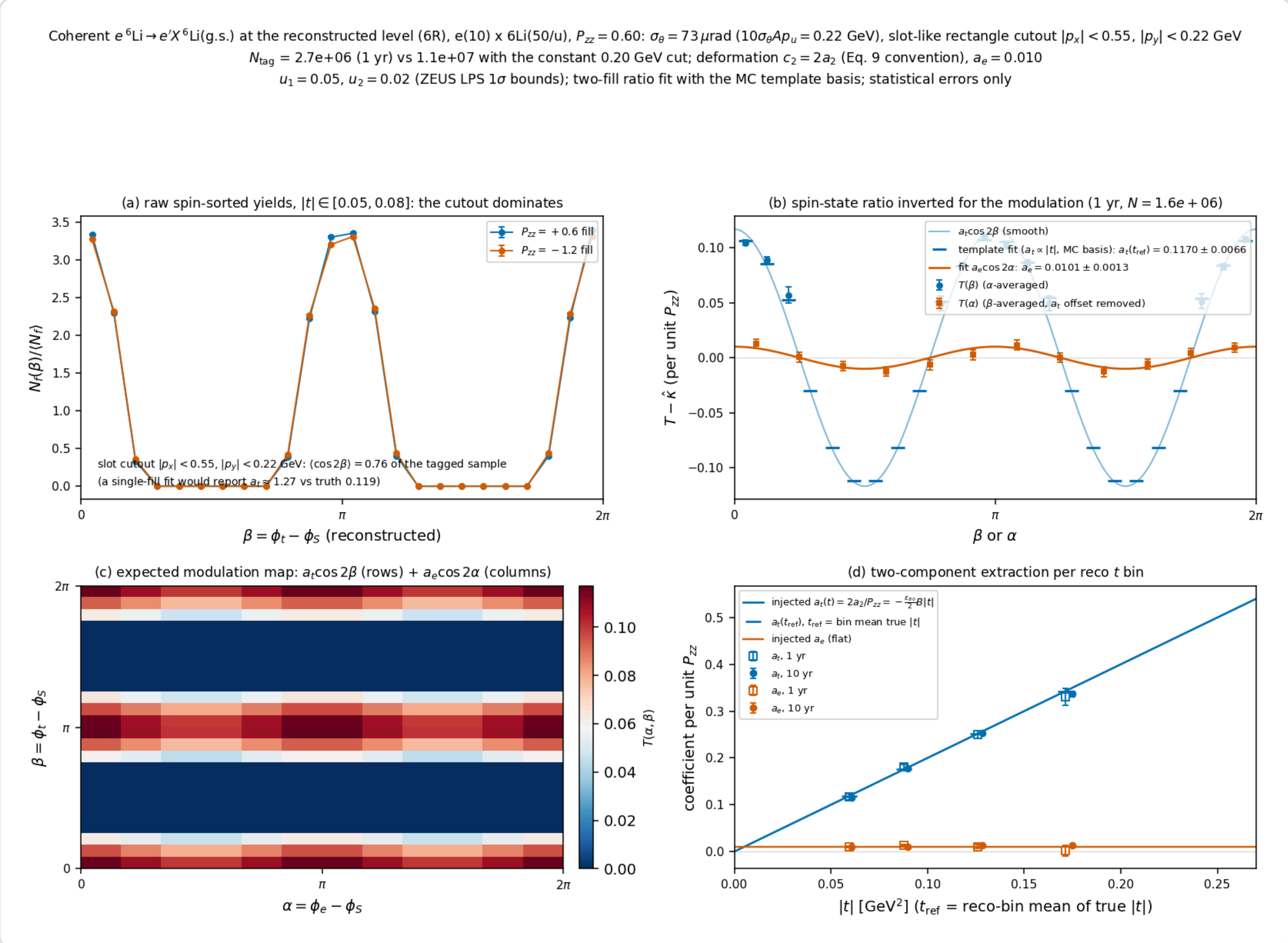


Money plot 7R — Δ(x,Q²) from reconstructed bins. xΔ points at the three sweet-spot Q^2 slices from the same pseudo-measurements, converted with the MC bin-centering factor K (model-derived, as in money plot 7, so the points sit on the injected curve by construction and carry the statistical errors of the reconstructed analysis); 1-year (open) and 10-year (filled) draws are independent.

Coherent channel (money plot 6R). The coherent response samples recoils from the exponential spectrum with $x_P \in [10^{-3}, 10^{-2}]$, builds the exact four-vector, and emulates the pots with the angular envelope $10\sigma_\theta \cdot 6p_u$ at σ_θ = 73 µrad (0.22 GeV at 50 GeV/u), a slot-like cutout $|p_x| < 0.55$, $|p_y| < 0.22$ GeV (the horizontal slot of the ePIC sensor planes [17], its width illustrative), and divergence smearing. The two-azimuth expected counts per fill are exact α-integrals per event of $1 + u_1\cos(\alpha-\beta) + u_2\cos 2(\alpha-\beta) + P_f[a_e\cos 2\alpha + a_t(t)\cos 2\beta + a_m\cos(\alpha+\beta)]$ with the *true* β and t inside and the reconstructed β and $|t| = p_T^2$ for the binning. Two lessons came out of this pass. First, the anchor convention: the full text of the deuteron paper fixes $d^2\sigma/d\Phi d|t| \propto 1 + 2\Sigma_n a_n e^{in\Phi}$ (its Eq. 9, with Φ the angle between the vector-meson momentum and the polarization axis in the γ*d frame), so the cos 2β coefficient is $2a_2$ — the deformation modulation is twice what money plot 6 injects ($c_2 = -(P_{zz}/2)\varepsilon_{BO}B|t|$ = 0.072 at ⟨|t|⟩ = 0.06 GeV² and P_{zz} = 0.6; `coherent.cos2phi_coefficient_deformation`). Second, the cutout makes the in-bin β distribution far from uniform (the acceptance varies by more than an order of magnitude around the circle) and correlates β with |t| inside a reco t bin (the blind directions hold the larger-|t| recoils), so the harmonic fit must use the acceptance-weighted basis from the response — ⟨cos 2β⟩, ⟨cos β⟩, ⟨sin β⟩ per bin, and for the t-dependent term the template $\langle(t/t_{ref})\cos 2\beta\rangle$ of the assumed linear shape — which is exactly the two-component fit $a_2(t) = c_{def}|t| + a_g$ of plans/06 in MC-response form. With the analytic (uniform-bin) basis the deformation coefficient is biased by 3–6% in the lowest t bin (1.25-aspect cutout); with the template it closes. A third, geometric lesson: the slot concentrates the tagged recoils near β ≈ 0 and π, where cos 2β hardly varies, so the lever arm on the deformation harmonic shrinks — its statistical error is three times that of a square cutout of similar acceptance (0.007 against 0.002 in the first |t| bin at one year), while the electron-azimuth term a_e is untouched. The pot geometry is therefore a design lever for the deformation measurement and irrelevant for the exotic-glue one.

RECO |T| BIN [GEV²]	N_{TAG}, 1 YR	$A_T(T_{REF})$ INJECTED	FIT, 1 YR	FIT, 10 YR	A_E FIT, 1 YR (INJ. 0.010)	A_E FIT, 10 YR
0.05–0.08	1.6×10⁶	0.119	0.117 ± 0.007	0.1161 ± 0.0021	0.0101 ± 0.0013	0.0100 ± 0.0004
0.08–0.12	7.4×10⁵	0.176	0.181 ± 0.005	0.1772 ± 0.0014	0.0130 ± 0.0018	0.0095 ± 0.0006
0.12–0.17	1.3×10⁵	0.252	0.250 ± 0.008	0.2533 ± 0.0024	0.0106 ± 0.0043	0.0137 ± 0.0014
0.17–0.25	1.8×10⁴	0.343	0.331 ± 0.018	0.337 ± 0.006	0.003 ± 0.012	0.013 ± 0.004

Table 3 — money plot 6R: template fit of the two-azimuth spin-state ratio per reconstructed |t| bin (coefficients per unit P_{zz}; ε_{BO} = −0.08, B = 50 GeV⁻²; u_1 = 0.05, u_2 = 0.02 at the ZEUS 1σ bounds [13]; N_{tag} = 2.7×10⁶ in year 1 for the angular envelope and the slot-like cutout, against 1.1×10⁷ for the constant 0.20 GeV cut). The cutout's fake ⟨cos 2β⟩ of the tagged sample is +0.77 (a single-fill fit would report $a_t \approx$ +1.3 for a true 0.12); the ratio removes it entirely.



Money plot 6R — the coherent channel at the reconstructed level. (a) Raw spin-sorted yields versus the reconstructed recoil azimuth β for the lowest |t| bin: the slot-like cutout shape dominates both fills identically (vertical recoils, β ≈ 0 and π, are the ones that clear the slot). **(b)** The modulation T(β) and T(α) recovered from the spin-state ratio (T(α) with the a_t offset removed), the template-fit prediction per β bin (ticks) and the smooth $a_t\cos 2\beta$ and $a_e\cos 2\alpha$ curves. **(c)** The expected (noise-free) modulation map in (α, β): rows from the deformation term, columns from the gluon-transversity term. **(d)** Extracted $a_t(t_{ref})$ and a_e per reco |t| bin (1 yr open, 10 yr filled) against the injected curves.

7.1 · Audit notes for this pass

Checked independently of the code that produced the numbers: the analytic error of the two-fill ratio, $2\sqrt{(2/N)}/(P_+-P_0)$/dilution, reproduces the pseudo-experiment spreads and the printed errors (spot 1: 1.21×10^{-4} for $N = 1.71\times10^{8}$); the exact expected counts reproduce the sampler's generator-level `effective_modulation` when binned in truth (test); the bin-centering factor returns the injected Δ exactly without noise (test); the sign of the deformation term is positive for $P_{zz} > 0$ and $\varepsilon_{B0} < 0$, consistent with the predicted sign flip relative to the deuteron; the Eq. (9) convention was read from the paper's text, not inferred; the relative-luminosity immunity is second order ($\delta\cdot\bar{P}/\sigma_P^2$ — a 2% offset moves the amplitude by 0.7%, a 10^{-4} offset by 3×10^{-5}); empty φ bins inside the cutout are handled with zero weight; the α-integrals of the coherent counts are analytic per event and the β smearing is absorbed in the MC basis, so the fitted coefficients refer to the unsmeared truth. Known limits of this pass, stated rather than hidden: the hadronic y is a parametrized 20% Gaussian (no ePIC performance document for the Σ method at y ≈ 0.01 was available, plans/04 #21); u1, u2 sit at the ZEUS 1σ bounds [13] and the slot dimensions are illustrative (#20); the φ′-efficiency harmonics are illustrative magnitudes; radiative corrections and backgrounds are not included; the 7R points are centered with the injected model (as in money plot 7); the template assumes the linear t-shape it fits, so a different shape is a model systematic to be varied.

References

[1] R. L. Jaffe, A. Manohar, *Nuclear gluonometry*, Phys. Lett. B 223 (1989) 218.

[2] P. Hoodbhoy, R. L. Jaffe, A. Manohar, *Novel effects in deep inelastic scattering from spin-one hadrons*, Nucl. Phys. B 312 (1989) 571.

[3] A. Bacchetta, M. Diehl, K. Goeke, A. Metz, P. J. Mulders, M. Schlegel, *Semi-inclusive deep inelastic scattering at small transverse momentum*, JHEP 02 (2007) 093, hep-ph/0611265 (covariant definitions of the azimuthal angles, Eqs. 2.4–2.5).

[4] A. Bacchetta, U. D'Alesio, M. Diehl, C. A. Miller, *Single-spin asymmetries: the Trento conventions*, Phys. Rev. D 70 (2004) 117504; M. Diehl, S. Sapeta, *On the analysis of lepton scattering on longitudinally or transversely polarized protons*, Eur. Phys. J. C 41 (2005) 515.

[5] F. Jacquet, A. Blondel, in *Proceedings of the Study of an ep Facility for Europe*, DESY 79/48 (1979) 391 (the hadronic-method kinematics).

[6] U. Bassler, G. Bernardi, *On the kinematic reconstruction of deep inelastic scattering at HERA: the Σ method*, Nucl. Instrum. Meth. A 361 (1995) 197.

[7] R. Abdul Khalek et al., *Science requirements and detector concepts for the Electron-Ion Collider: EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447 (kinematic reconstruction methods, far-forward detectors, 10σ Roman-Pot cut, beam parameters).

[8] A. Jentsch, Z. Tu, C. Weiss, *Deep-inelastic electron-deuteron scattering with spectator nucleon tagging at the future Electron-Ion Collider: extracting free nucleon structure*, Phys. Rev. C 104 (2021) 065205, arXiv:2108.08314 (far-forward acceptance and resolution model).

[9] H. Mäntysaari, F. Salazar, B. Schenke, C. Shen, W. Zhao, *Spatial imaging of polarized deuterons at the Electron-Ion Collider*, Phys. Lett. B 858 (2024) 139053, arXiv:2408.13213 (the recoil-azimuth modulation anchoring the deformation term).

[10] W. Chang, E.-C. Aschenauer, A. Jentsch, A. Kumar, Z. Tu, Z. Yin, *Opportunities for imaging light nuclei with a second interaction region at the Electron-Ion Collider*, Phys. Rev. D 113 (2026) 032018, arXiv:2511.05638 (IR-8 secondary focus, intact-recoil tagging).

[11] This repository: `reports/cos2phi_money_plots_report` (projections and assumptions), `plans/06` (coherent background budget), `plans/07` (letter plan, WP3), `evgen/polligen/reco.py`, `evgen/polligen/recopseudo.py` and the scripts named in the text (this note's computations).

[12] W. Cosyn, B. Roldan Tomei, A. Sosa, A. Zec, *Polarization options in inclusive DIS off tensor polarized deuteron*, Eur. Phys. J. A 61 (2025) 83, arXiv:2410.12764 (exact spin-1 decomposition, Eqs. 10, 17, 20).

[13] ZEUS Collaboration, S. Chekanov et al., *Deep inelastic scattering with leading protons or large rapidity gaps at HERA*, Nucl. Phys. B 816 (2009) 1, arXiv:0812.2003 (leading-proton azimuthal asymmetries A_{LT}, A_{TT}; LPS resolutions; beam transverse-momentum spread).

[14] N. N. Nikolaev, A. V. Pronyaev, B. G. Zakharov, *Azimuthal asymmetry as a new handle on σ_L/σ_T in diffractive DIS*, arXiv:hep-ph/9812212 (Eq. 1: the y-dependence of the cos φ and cos 2φ terms).

[15] A. V. Pronyaev, *The forward cone and L/T separation in diffractive DIS*, arXiv:hep-ph/9808432 (Eq. 7: the LT/T ratio).

[16] M. Ruspa (H1 and ZEUS), *Inclusive diffraction at HERA*, arXiv:hep-ex/0206031.

[17] A. Jentsch (ePIC), *Far-Forward Detectors and Physics with ePIC @ the EIC*, talk at DIS 2023, East Lansing, 27–31 March 2023 (far-forward acceptance table, Roman-Pot geometry, tracker p_T resolution with and without beam effects). Local copies of [9], [12]–[17] are indexed in `refs/README.md`.

Conventions: head-on frame with the ion along +z and the electron along −z; θ′ is the scattered-electron angle from the electron beam; per-nucleon (x, y, s) throughout; whole-nucleus four-momenta where a mass matters. Beam divergences are the proton values implied by the documented Roman-Pot cuts (73 and 149 μrad); the lithium optics are undocumented (plans/04 #11). Hadronic-method resolutions are assumptions (15–30%), not ePIC results (plans/04 #21); u_1, u_2 sit at the ZEUS 1σ bounds; the slot dimensions of the Roman-Pot cutout are illustrative (#20). Built by `reports/build_report.py`; figures from the `evgen/scripts` named in the text; display mathematics typeset at build time.